

MCAT BrainLift

Owners

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Purpose

To understand how MCAT students build exam-ready performance and identify where current study tools (including Anki) fail to bridge the gap between studying and actual test-day performance.

In Scope

- How the MCAT is structured, scored, and what it actually tests
- What the current MCAT study process looks like end-to-end (tools, order, integration)
- Where current study methods fail to bridge recall to application to exam performance
- What readiness actually means for MCAT specifically

Out of Scope

- Generic study tips / productivity advice
- AI/LLM architecture for card generation
- Deep dives into specific MCAT content (e.g. the Krebs cycle, amino acid structures, etc.)

DOK 4: Spiky Points of View (SPOVs)

- **Spiky POV 1: Readiness is not an end state; it's a continuous signal that current tools fail to surface until it's too late.**
 - Under the standard MCAT prep flow (content review → practice problems → full-length practice tests in the last weeks), students spend months doing content review without any readiness signal, then discover their actual preparation level only in the final weeks before the exam. Students spend on average 240 hours over 3 months before they have a basic readiness signal: by the time a student discovers that their preparation hasn't been good enough, it's too late to change their process. Readiness measurement needs to move from the finish line to a live signal updated throughout the learning process.
- **Spiky POV 2: Students assume that pairing content review with practice questions closes the gap between knowing and performing. In reality, the gap between them is exactly where MCAT performance is lost.**
 - Traditionally, students pair Anki-based content review with UWorld-style practice questions under the assumption that the two together cover the full spectrum of what the exam tests: one builds the knowledge base, while the other trains application. This assumption misses that declarative knowledge (the basic facts) and procedural knowledge (how to apply them) are distinct cognitive outcomes

that don't automatically transfer between each other: knowing a fact in the context you studied it in doesn't mean you can directly deploy it to an unfamiliar passage. Closing this gap requires a tool that integrates memory tracking and performance measurement into a single feedback loop instead of treating them as disjoint layers.

- **Spiky POV 3: Current tools record what students got wrong, but not why. Without that distinction, students have no consolidated view of what's actually breaking down -- and neither does the model.**
 - A wrong answer caused by misreading the question, a reasoning error, or a genuine content gap all carry completely different information. However, current tools don't capture the difference. Existing models reason as follows: "You missed an electrochemistry question → do more electrochemistry". A stronger model with error typing data instead suggests "Your electrochemistry content recall is fine. Your misses are mostly passage-to-equation mapping errors under timed conditions. Do 10 targeted passage-mapping drills, not another content review chapter." Current tools personalize at the level of topics and schedules, but not at the level of the individual learner. Tools need to capture not just what students got wrong, but why -- without that data, meaningful personalization is impossible.

Experts

- **Expert 1: Robert A. Bjork**
 - **Who:** Distinguished Research Professor of Psychology at UCLA; co-founder of the Bjork Learning and Forgetting Lab.
 - **Focus:** Human learning, memory, and "desirable difficulties." Known for research on interleaving, spacing effects, and the gap between how people think they learn vs. how they actually learn.
 - **Why Follow:** His 2008 paper with Kornell on blocked vs. interleaved practice is the direct academic anchor for Subcategory 3.3, and his broader work on desirable difficulties explains why spaced repetition optimizes for short-term recall rather than flexible, transferable knowledge.
 - **Where:** <https://www.psych.ucla.edu/faculty-page/bjork/> | <https://bjorklab.psych.ucla.edu/>
- **Expert 2: John D. Bransford**
 - **Who:** Shauna C. Larson Endowed Chair of Learning Sciences at the University of Washington; former Co-Chair of the National Academy of Sciences committee on learning.
 - **Focus:** Human cognition, learning transfer, and the design of effective learning environments. Lead editor of *How People Learn* (National Academies, 2000).
 - **Why Follow:** *How People Learn* is the foundational source for the recall vs. application gap in Category 3. His argument that usable knowledge is not a list of disconnected facts is the theoretical basis for why Speedrun's three scores need to be separate.
 - **Where:** <https://www.nationalacademies.org/read/9853>
- **Expert 3: Emil Chuck, PhD**

- **Who:** Director of Advising Services at HPSA; author at Student Doctor Network Review.
 - **Focus:** MCAT preparation trends and analysis of AAMC Post-MCAT Questionnaire data.
 - **Why Follow:** His 2025 SDN Review article is the primary empirical source for Category 2, documenting the shift toward flashcard-based passive study and the decline of active-learning behaviors among test-takers.
 - **Where:** <https://www.studentdoctor.net/author/emilchuck/>
- **Expert 4: Anthony Artino Jr., PhD**
 - **Who:** Professor of health professions education, currently at the University of Ottawa.
 - **Focus:** Metacognition, self-regulated learning, and calibration bias in medical students.
 - **Why Follow:** His 2019 study showing that 95-98% of medical students overestimate their own performance is the academic anchor for Category 4.2 and the basis for why Speedrun needs externally calibrated scores.
 - **Where:** <https://pmc.ncbi.nlm.nih.gov/articles/PMC6775028/>

DOK 3: Insights

From Category 1: MCAT Structure

- **Insight 1:** The MCAT is a reasoning test that requires content knowledge as a prerequisite, not a content test that sometimes requires reasoning. The Skill 1 figure (~35% of questions require direct content knowledge) means that students who optimize solely for recall are prepping well for a minority of the exam but ignoring the remaining 65%. The most prevalent framing of “learn the content first, then practice) is backward.
- **Insight 2:** The CARS category is a large blindspot for how students think about MCAT prep: because it tests zero content, strong CARS performance can not be trained through content study of any kind. Tools that only address content (including Anki) are irrelevant to ~23% of the exam.

From Category 2: Current Study Process

- **Insight 3:** The rise of Anki along with the decline of self-explanation and answer review suggests that students aren’t adding Anki as a supplement to good habits; instead, they’re replacing them with it. The result is more time spent on recall (which isn’t bad by itself), but comes at the cost of active processing that builds critical thinking ability.
- **Insight 4:** The 3 phase prep funnel (content → question banks → full length practices) treats readiness as something that emerges at the end of prep rather than something measurable during the process. As a result, students spend months in phase 1 and 2 without a good gauge of whether they’re actually on track, then discover their actual readiness only in the final weeks before.

From Category 3: Where Methods Break Down

- **Insight 5:** The same poor studying behavior manifests differently across score ranges: at 505 - 510, most errors are content gaps, whereas reasoning is the primary issue at 510+. Tools that diagnose all students the same way fail to accurately identify the real bottleneck.
- **Insight 6:** Studying the MCAT in blocks gives students false confidence relative to actual ability: it feels productive because familiarity accumulates quickly, but interleaving on the real test exposes substantial gaps in ability. As a result, students are getting strong positive feedback from formats that prepare them poorly for the actual test.

From Category 4: What Readiness Actually Means

- **Insight 7:** Practice test scores (done only after months of prep) are the only true readiness signal students currently have, but they don't measure the quality of preparation leading up to it. A student can reach a 510 on a full-length through pattern recognition without having the underlying skills to perform consistently.
- **Insight 8:** Students gauge readiness based on what they can easily measure through low-hanging metrics like Anki card completion. However, self-assessment is least accurate for complex tasks like passage reasoning (which is exactly what determines their actual score). Students feel false readiness because they're selecting for metrics they do well in, not because they're truly exam-ready.

DOK 2: Knowledge Tree

Category 1: MCAT Structure & What It Actually Tests

- **Subcategory 1.1: Exam format, sections, and scoring**
 - **Source 1:** AAMC, "What's on the MCAT Exam?"
 - **DOK 1 - Facts**
 - There are four sections on the MCAT: (1) Chemical and Physical Foundations of Biological Systems; (2) Critical Analysis and Reasoning Skills (CARS); (3) Biological and Biochemical Foundations of Living Systems; (4) Psychological, Social, and Biological Foundations of Behavior.
 - Sections 1, 3, and 4 are science sections with 59 questions each. Section 2 (CARS) has 53 questions, for a total of 230 questions.
 - Each science section is allotted 95 minutes; CARS is 90 minutes. Total testing time is 6 hours 15 minutes; total test-day time is 7.5 hours.
 - Each science section combines passage-based question sets (~10 passages) with ~15 standalone questions. CARS is entirely passage-based (~9 passages) with no discrete questions and no subject-matter knowledge required.

- The science sections test four skills (SIRS): Skill 1 (Knowledge of Scientific Concepts and Principles), Skill 2 (Scientific Reasoning and Problem-Solving), Skill 3 (Reasoning About Research Design), Skill 4 (Data-Based and Statistical Reasoning). CARS tests three skills: Foundations of Comprehension, Reasoning Within the Text, and Reasoning Beyond the Text.
- **Summary & Analysis**
- The MCAT is a 230-question, 7.5-hour exam across four sections. Within the science sections, the majority of questions are passage-based, meaning students have to apply knowledge to new contexts rather than retrieve facts in isolation. The skills framework makes clear the exam is designed to test reasoning and problem-solving, not just content recall.
- **Link to Source:**
<https://students-residents.aamc.org/prepare-mcat-exam/whats-mcat-exam-pdf-outline>
- **Subcategory 1.2: What the test rewards (application vs. recall)**
 - **Source 1:** AAMC, "Scientific Inquiry & Reasoning Skills: Overview"
 - **DOK 1 - Facts**
 - Skill 1 (Knowledge of Scientific Concepts and Principles) asks students to recognize, recall, or define concepts -- the closest analog to flashcard-style recall.
 - Skill 2 (Scientific Reasoning and Problem-Solving) asks students to apply knowledge to novel situations, combine multiple content areas, and make predictions, going beyond recall.
 - Skill 3 (Reasoning About Research Design) asks students to evaluate experimental methods and extrapolate conclusions from study design.
 - Across the three science sections, only ~35% of questions fall under Skill 1; the remaining ~65% require Skills 2-4.
 - AAMC's framing: the exam tests the ability to "combine scientific knowledge with skills in scientific inquiry and reasoning," not content recall alone.
 - **Summary & Analysis**
 - The SIRS framework shows the core problem with Anki-only prep: flashcards are built for Skill 1, but Skill 1 is a minority of the exam. Most questions require students to do something with knowledge -- apply it to a novel passage, evaluate an experiment, or interpret data. Even Skill 1 questions are typically passage-embedded, so isolated fact recall is rarely sufficient.
 - **Link to Source:**
<https://students-residents.aamc.org/whats-mcat-exam/scientific-inquiry-reasoning-skills-overview>

Category 2: The Current MCAT Study Process

- **Subcategory 2.1: How students typically prepare (tools, timelines, order)**
 - **Source 1:** SDN Review, "MCAT Preparation Trends" (Emil Chuck, PhD, 2025)
 - **DOK 1 - Facts**

- The standard prep workflow follows three phases: content review (books, videos, Anki), practice questions (UWorld, third-party QBanks), and AAMC full-lengths in the final stretch.
- Over 90% of test-takers accessed AAMC full-length practice exams; other materials used include unscored practice exams (65.9%), question packs (63.1%), Section Bank (61.2%), and CARS diagnostic (48.3%).
- Only 31.7% enrolled in a structured test-prep course; self-directed study is now the dominant approach.
- Use of free online flashcard programs jumped from 37.3% to 50.2% between 2020 and 2024; flashcard use for memorization overall rose from 67.4% to 71.4%.
- Reviewing answer explanations declined (74.7% to 70.9%) and self-explanation declined (56.5% to 52.0%), pointing to a shift toward passive self-quizzing.
- **Summary & Analysis**
- The current MCAT prep workflow is self-directed and anchored on AAMC official materials. Students converge on a three-phase funnel regardless of timeline: build content, apply through practice questions, calibrate against AAMC exams. Notably, flashcard and passive self-quizzing use is rising while deeper active-learning behaviors like self-explanation and answer review are declining, meaning students are leaning into tools that optimize for recall even as the exam rewards reasoning.
- **Link to Source:**
<https://www.studentdoctor.net/2025/12/04/mcat-preparation-trends-insight-s-from-the-aamc-post-mcat-questionnaire-and-the-hpsa-2024-2025-application-experience-surveys/>

- **Source 2:** AAMC, "Four Medical Student Tips for Preparing for the MCAT Exam"
 - **DOK 1 - Facts**
 - AAMC survey data shows students prepare on average 20 hours per week for three months, totaling roughly 240 hours, though prep ranges widely from 3 to 6 months depending on commitments and target score.
 - Above-average scorers typically invest 300+ hours; competitive scorers aiming for 515+ often study 400-500 hours.
 - AAMC explicitly advises spreading study out rather than cramming, noting spaced practice locks information into long-term memory more effectively.
 - **Summary & Analysis**
 - The 3-month/240-hour figure is the AAMC's own average, but it is a floor not a ceiling -- students targeting competitive scores routinely invest significantly more. AAMC's guidance prioritizes practice-exam-driven iteration over linear content review, reinforcing that application under test conditions is the primary mechanism of improvement, not content accumulation alone.
 - **Link to Source:**
<https://students-residents.aamc.org/premed-navigator/four-medical-student-tips-preparing-mcat-exam>

- **Subcategory 2.2: How Anki fits into current prep workflows**

- **Source 1:** Jack Westin, "How to Use Anki Effectively for the MCAT" (2026)
 - **DOK 1 - Facts**

- The two most widely used premade decks are MileDown (~2,900 cards) and AnKing MCAT (~6,256 cards); students typically do 50-100 new cards per day during dedicated prep.
- Anki is used primarily during the content review phase, running alongside books and videos, with daily reviews maintained throughout prep.
- Anki works best for recall-heavy sections (Psych/Soc and Biochemistry) and has near-zero utility for CARS, which requires reasoning skills that flashcards don't train.
- When used as a retention tool alongside practice questions, Anki is high-ROI; when used as a primary content source, it "feels productive but does not translate into passage performance."
- Anki's scheduling algorithm (SM-2, now superseded by FSRS in Anki 23.10) optimizes recall, not application or reasoning.
- **Summary & Analysis**
- Anki has a well-defined but limited role in MCAT prep: it is a retention tool for recall-heavy content, not a comprehension or reasoning tool. Students who treat it as a supplement to practice-based study use it well; students who treat it as their primary resource hit a ceiling of high card completion rates but low passage performance. CARS is the clearest illustration of this -- the section that consistently produces the lowest scores is one that spaced repetition cannot address at all.
- **Link to Source:**
<https://jackwestin.com/blog/how-to-use-anki-effectively-for-the-mcat/>

Category 3: Where Current Methods Break Down

- **Subcategory 3.1: The recall to application gap**
 - **Source 1:** Bransford, Brown & Cocking, "How People Learn" (National Academies, 2000)
 - **DOK 1 - Facts**
 - "Usable knowledge is not the same as a mere list of disconnected facts" -- transfer of learning requires understanding core principles, not just memorizing facts.
 - Transfer is directly affected by the degree to which students learn with understanding rather than memorize sets of facts or follow fixed procedures.
 - Assessment focusing on recall rather than use actively discourages transfer -- students optimized for recall tests are not being trained for application.
 - Bjork & Bjork (1992), cited in Bransford: repeating the same problem types mainly builds recall, not adaptable knowledge that can be applied flexibly in new situations.
 - **Summary & Analysis**
 - The cognitive science on learning transfer establishes that recall and application are distinct outcomes. A student can pass a recall test and fail a transfer test on the same material, and study tools that only practice recall build a necessary but not sufficient condition for MCAT performance. For Speedrun, this is the theoretical basis for measuring memory score and performance score separately rather than combining them into one number.

- **Link to Source:** <https://www.nationalacademies.org/read/9853/chapter/6>
- **Source 2:** AMEE Guide No. 176, "Teaching for Transfer of Learning in Health Professions Education" (2024)
 - **DOK 1 - Facts**
 - Declarative knowledge = "knowing what" (facts, definitions, concepts).
Procedural knowledge = "knowing how" (applying facts to solve problems, make predictions, answer clinical questions).
 - Both types are essential in medical assessment, but they require different assessment methods and cannot be measured by the same tool.
 - Students need solid declarative knowledge as a foundation before procedural knowledge can be built, but declarative knowledge alone does not produce procedural competence.
 - "Blunt short-term memorization of declarative knowledge neither results in actual learning nor clinically-useful application."
 - **Summary & Analysis**
 - The AMEE framework maps directly onto Speedrun's three-score model: memory score measures declarative knowledge, performance score measures procedural knowledge. These are not on the same continuum and require separate measurement. This is also the academic basis for why Anki alone is insufficient -- it is a declarative knowledge tool in an exam that increasingly rewards procedural knowledge above the 505 score level.
 - **Link to Source:**
<https://www.tandfonline.com/doi/full/10.1080/0142159X.2024.2414823>
- **Subcategory 3.2: Why passage-based questions expose memory limits**
 - **Source 1:** Premier MCAT Prep, "MCAT Score Not Improving" (2026)
 - **DOK 1 - Facts**
 - At the 505 level, content gaps account for roughly 35% of missed questions; by 510+, content gaps drop to roughly 15-20%.
 - Above 510, missed questions rarely involve pure content recall -- they require interpreting passage data through the lens of content knowledge.
 - Timing problems are often content automaticity problems in disguise: spending 30 seconds reconstructing a concept leaves 30 fewer seconds for reasoning.
 - **Summary & Analysis**
 - The relationship between content knowledge and MCAT score is nonlinear. Content gaps dominate at lower score ranges but become a minority of errors as students improve, so a tool that only tracks memory retention accurately diagnoses low scorers but consistently misdiagnoses higher scorers. For Speedrun, this also points to a specific goal: content knowledge needs to be fast enough that it does not consume working memory during passage reasoning, which is a different goal than simple recall.
 - **Link to Source:**
<https://premiermcatsprep.com/blog/mcat-score-not-improving>
- **Subcategory 3.3: Cross-domain synthesis failures**
 - **Source 1:** Kornell & Bjork, "Learning Concepts and Categories: Is Spacing the Enemy of Induction?" (Psychological Science, 2008)

- **DOK 1 - Facts**
- Blocked practice (studying one category at a time) produced significantly worse transfer to novel items than interleaved practice, even though participants consistently believed blocking was more effective.
- Interleaving forces discrimination between categories rather than accumulation within one, which is the mechanism underlying inductive learning and novel context performance.
- Students trained with blocked sequences were unprepared for interleaved test conditions; those trained with interleaved sequences were unaffected by testing conditions, meaning blocked learning produces brittle, context-dependent knowledge.
- In medical education, interleaved practice on ECG interpretation produced 46% diagnostic accuracy vs. 30% for blocked practice (McSparron et al., 2019).
- **Summary & Analysis**
- The MCAT is structurally an interleaved test, with a single passage routinely requiring biochemistry, experimental reasoning, and data interpretation simultaneously. Students who study primarily by subject in isolation are training in blocked format and are underprepared for this cross-domain demand. The metacognitive illusion Kornell and Bjork identify makes this hard to self-diagnose: studying one subject at a time feels effective because familiarity accumulates quickly, but this feeling is decoupled from actual transfer performance.
- **Link to Source:**
https://www.researchgate.net/publication/5277058_Learning_Concepts_and_Categories_Is_Spacing_the_Energy_of_Induction

Category 4: What Readiness Actually Means for MCAT

- **Subcategory 4.1: How students currently gauge readiness**
 - **Source 1:** Premier MCAT Prep, "MCAT Practice Test Scores vs. Actual" + MCAT.tools Score Predictor data
 - **DOK 1 - Facts**
 - AAMC full-length practice exams are the most accurate predictor of real MCAT score -- the community consensus is to judge readiness on AAMC data, not third-party tests.
 - Third-party practice tests vary considerably: Kaplan deflates by ~10 points, Princeton Review by ~15 points, Blueprint by ~5 points, meaning a student's perceived readiness shifts depending on which tests they take.
 - MCAT.tools score predictor, based on 3,560 data points, has a standard error of +/- 3.3 points -- 68% of students fall within 3.3 points of their predicted score.
 - Current community standard: a student is considered ready when their last 2-3 AAMC full-lengths are at or above their target score with no chronically weak section.
 - **Summary & Analysis**
 - Current MCAT readiness assessment is entirely practice-test-driven. Students triangulate a handful of AAMC full-lengths and make a go/no-go decision, which is the best available signal but a lagging indicator -- students only discover they were not ready after investing 300-500 hours

of prep. The third-party deflation problem also shows that students regularly receive inaccurate readiness signals, and there is no standardized framework for what readiness means before a student starts taking full-lengths.

- **Link to Source:**

<https://premiermcatsprep.com/blog/mcat-practice-test-scores-vs-actual> +
<https://mcat.tools/mcat-scores-calculator-predictor>

- **Subcategory 4.2: The gap between perceived and actual readiness**

- **Source 1:** Artino et al., "First-year medical students' calibration bias and accuracy across clinical reasoning activities" (Advances in Health Sciences Education, 2019)

- **DOK 1 - Facts**

- Study of 157 first-year medical students found that 98% overestimated their performance on history-taking tasks and 95% overestimated on physical exam tasks.
 - Poor calibration is especially pronounced in lower-performing students (Dunning-Kruger effect) -- the students who most need accurate readiness feedback are the least able to generate it themselves.
 - Calibration accuracy improves when self-assessment is broken into specific subtasks rather than global judgments, meaning granular feedback produces better self-awareness than holistic scores.
 - Pre-examination self-assessment accuracy was very low ($r = 0.042$ to 0.243) in a study of medical students predicting oral exam performance.

- **Summary & Analysis**

- Research on medical student calibration shows that self-assessed readiness is systematically unreliable, with students -- especially lower-performing ones -- consistently overestimating how prepared they are. This is the basis for why Speedrun needs to provide externally measured, calibrated scores rather than relying on students' own sense of readiness. Showing separate memory, performance, and readiness scores is also more likely to produce accurate self-assessment than a single blended number, because it forces students to confront specific gaps rather than averaging them away.
 - **Link to Source:** <https://pmc.ncbi.nlm.nih.gov/articles/PMC6775028/>